FragTree — Final Clustering Report

Generated: 2026-04-10 12:36:26 | min_cluster_size=69 | min_samples=67

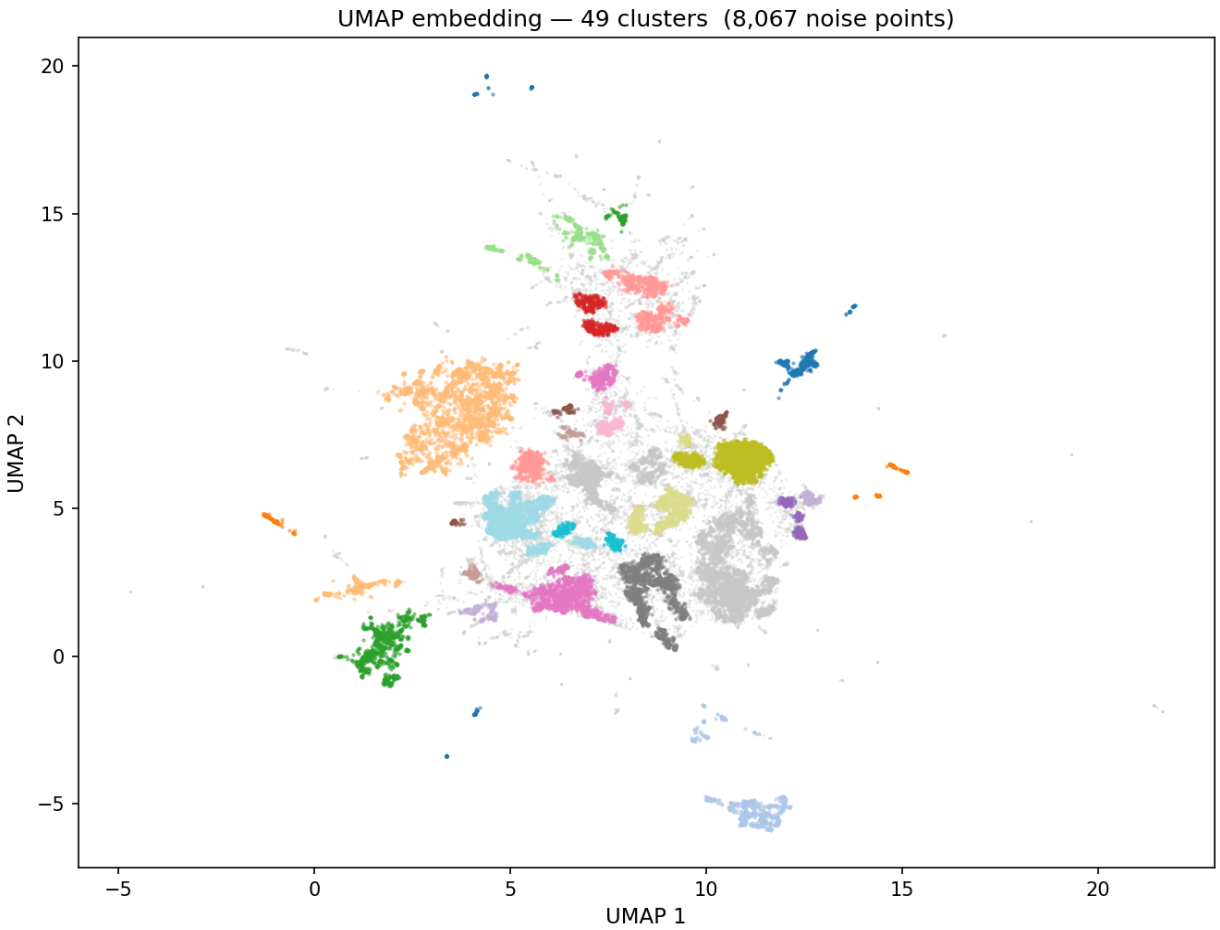
Global Clustering Metrics

DBCV is the primary metric. ARI and AMI use CAS number groupings as soft ground truth. BSS/TSS measures the fraction of total variance explained by the cluster structure (higher = better).

Metric	Value	Target	Notes
DBCV (primary)	0.4348	Maximize	Density-based clustering validation
Silhouette score	0.4678	Maximize	Mean intra vs inter-cluster cohesion
Noise fraction	0.265	< 0.30	8,066 points labelled noise
BSS / TSS	0.7116	Maximize	Fraction of variance explained by cluster structure
Mean persistence	0.1606	Maximize	Stability across density thresholds
N clusters	49	-	Noise cluster (-1) not counted
ARI	0.0000	1.0 ideal	Agreement with CAS number groupings
AMI	-0.0000	1.0 ideal	ARI normalised for cluster size distribution

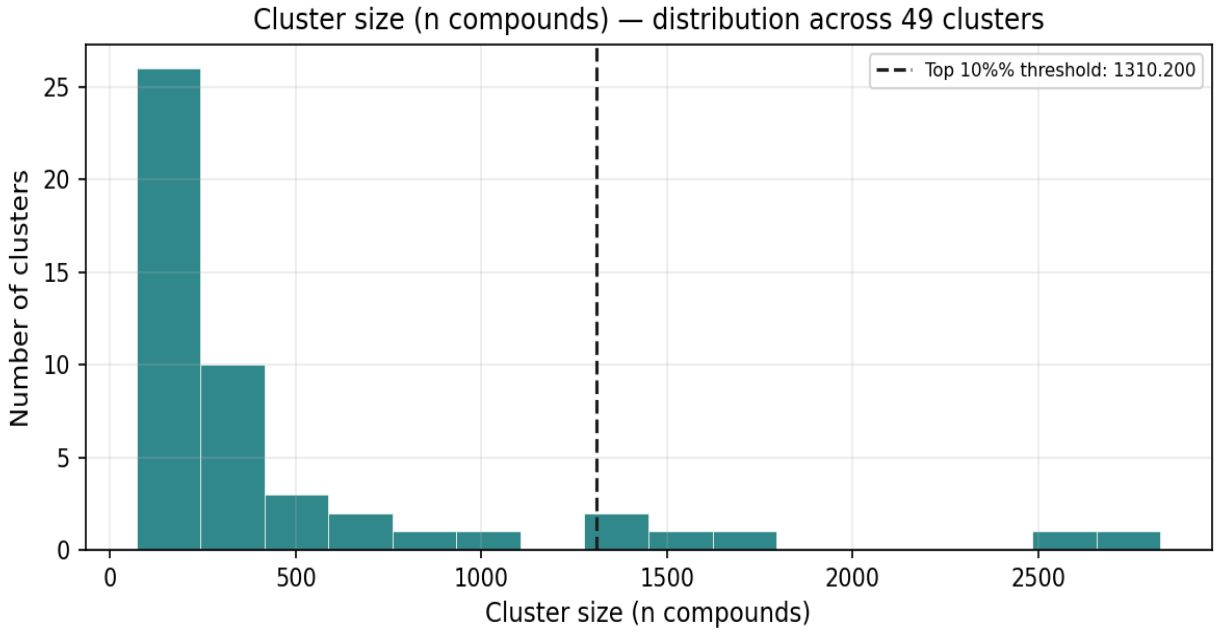
UMAP Embedding — Coloured by Cluster

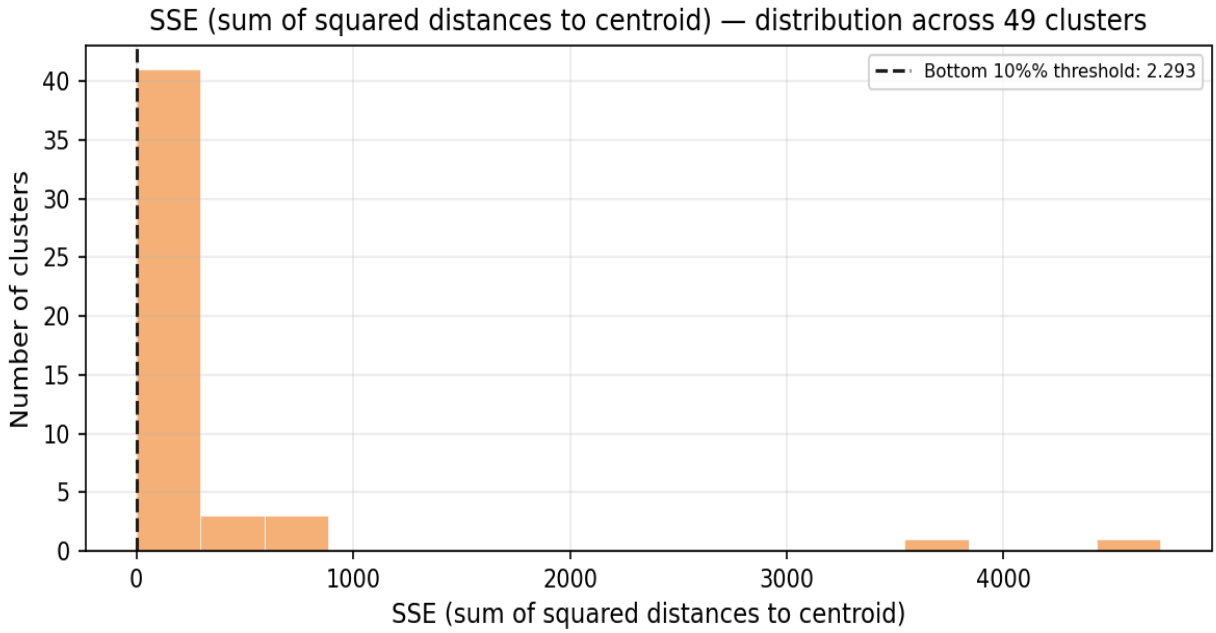
Each point is one compound. Grey = noise (label -1).

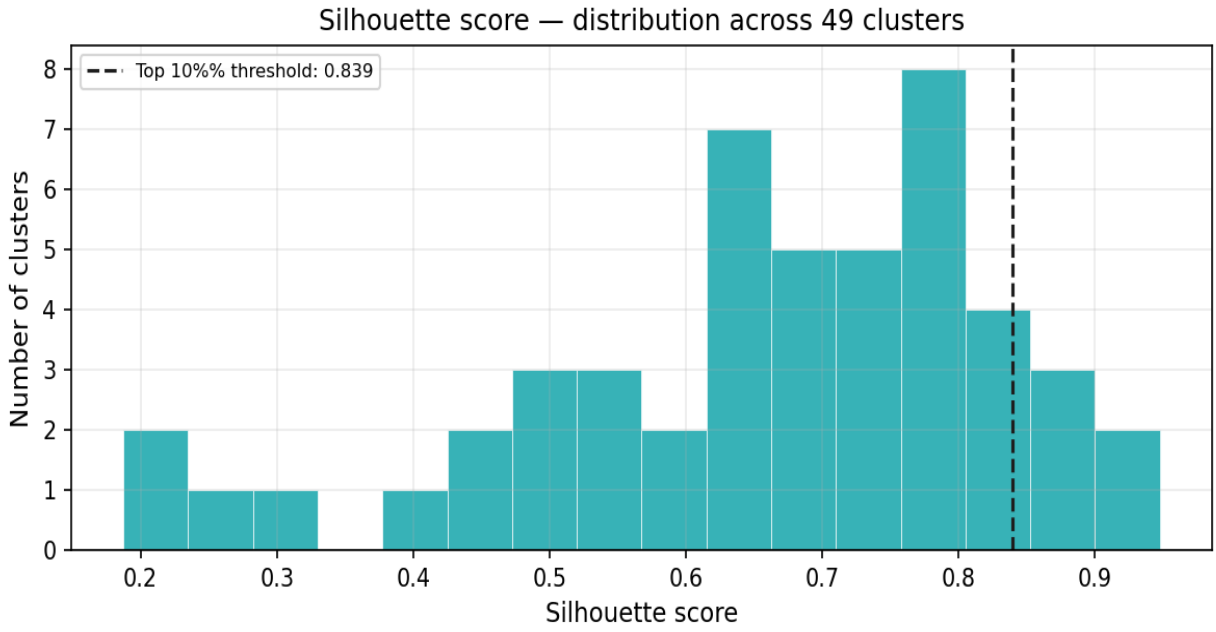

UMAP embedding — 49 clusters (8,067 noise points)
UMAP 2
UMAP 1
20
15
10
5
0
−5
−5
0
5
10
15
20

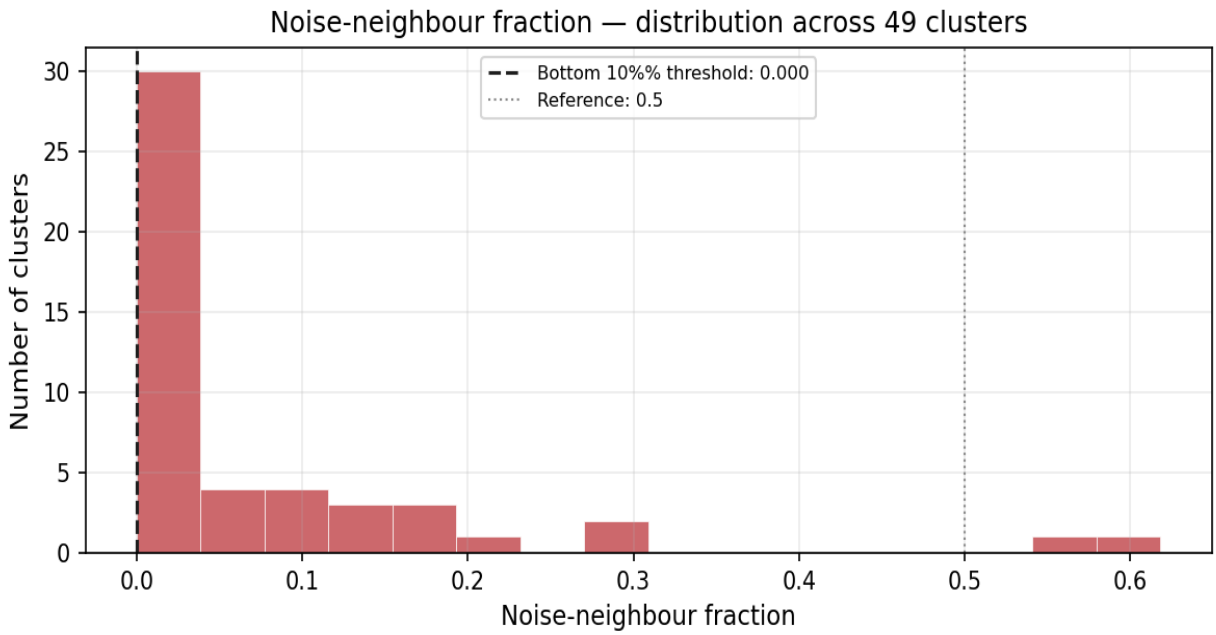
Per-Cluster Metric Distributions

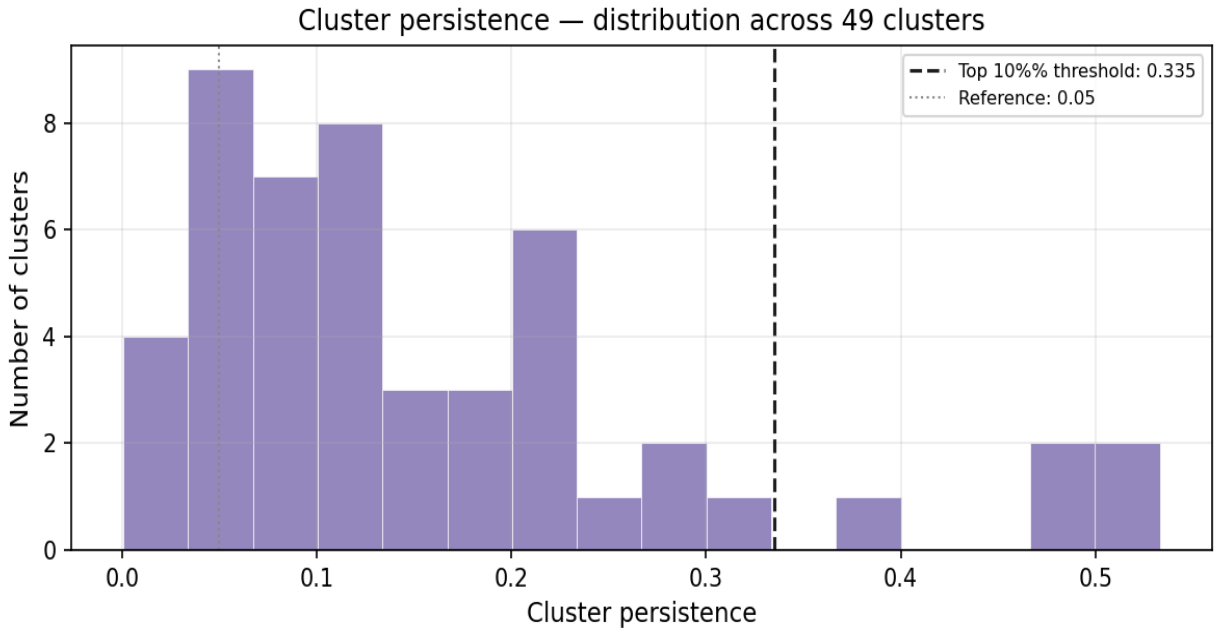
Each histogram shows how a metric is distributed across all clusters. Ideal training clusters sit in the right tail of size, silhouette, and persistence, and the left tail of SSE and noise-neighbour fraction.






Silhouette score — distribution across 49 clusters
Top 10%% threshold: 0.839
Number of clusters
Silhouette score
0
1
2
3
4
5
6
7
8
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9


Noise-neighbour fraction — distribution across 49 clusters
Bottom 10%% threshold: 0.000
Reference: 0.5
Number of clusters
Noise-neighbour fraction
0
5
10
15
20
25
30
0.0
0.1
0.2
0.3
0.4
0.5
0.6


Cluster persistence — distribution across 49 clusters
Top 10%% threshold: 0.335
Reference: 0.05
Number of clusters
Cluster persistence
0
2
4
6
8
0.0
0.1
0.2
0.3
0.4
0.5

Top-10% Cluster Summary Table

A cluster is included if it falls in the top 10% of ANY per-cluster metric (or bottom 10% for SSE and noise-neighbour fraction where lower is better). Sorted by silhouette score descending. Use this table to identify the best candidate cluster for model training.

Cluster	Size	SSE	Silhouette	Noise nbr frac	Persistence
25	82	0.46	0.9476	0.061	0.4737
5	163	9.32	0.9200	0.110	0.5184
13	108	1.15	0.8981	0.093	0.4820
0	77	29.95	0.8689	0.000	0.0192
26	180	3.02	0.8629	0.172	0.5331
35	155	9.53	0.8334	0.000	0.0624
6	96	4.13	0.8060	0.000	0.1092
41	86	1.07	0.7975	0.000	0.0406
24	101	2.36	0.7848	0.000	0.1519
1	74	45.93	0.7796	0.000	0.0095
7	92	6.21	0.7764	0.000	0.0351
2	416	179.19	0.7736	0.091	0.3774
30	71	2.02	0.7727	0.000	0.0007
21	140	1.80	0.7661	0.000	0.2662
27	105	2.87	0.7548	0.000	0.0773
47	175	6.49	0.7243	0.000	0.0868
22	198	4.23	0.6943	0.000	0.2199
44	140	5.00	0.6855	0.000	0.0551
46	126	3.34	0.6691	0.000	0.0549
45	140	4.68	0.6648	0.000	0.0576
16	249	14.21	0.6600	0.000	0.0894
23	201	17.43	0.6471	0.000	0.1175
29	349	36.37	0.6275	0.000	0.1442
33	106	5.37	0.6260	0.000	0.0081
18	374	65.34	0.6151	0.000	0.0591
32	300	15.77	0.5908	0.000	0.1769
20	225	18.69	0.5906	0.000	0.1960

40	275	16.34	0.5522	0.000	0.1032
17	478	109.79	0.5112	0.000	0.0848
12	308	75.57	0.4906	0.000	0.0945
39	1,634	386.41	0.4780	0.016	0.2233
43	646	122.56	0.4573	0.000	0.1150
31	1,298	517.10	0.3977	0.000	0.1260
9	2,829	4723.40	0.3018	0.307	0.1079
34	1,359	795.98	0.2386	0.055	0.0741
48	1,539	621.76	0.2303	0.000	0.0564
36	2,605	3575.92	0.1874	0.025	0.0510